CLAIMS

What is claimed is:

1. A method for transmitting data across a parallel interface, comprising the steps of:

- providing a parallel interface comprising a plurality of channels, each channel capable of representing one of a set of discrete states;
- encoding a numerical value as a set of a plurality of parameters, wherein said parameters provide instructions to generate the numerical value from a mathematical series; and
- mapping the entire set of said parameters to a single discrete state of a single channel of the parallel interface, thereby creating an effective data bus width that is a multiple of the number of physical channels.

2. The method of claim 1, wherein the effective data bus width is greater than 1,000,000 bits.

3. The method of claim 1, wherein the effective data bus width is greater than 1 gigabit.

4. The method of claim 1, wherein the set of parameters comprises a series identifier and a position within the series.

5. The method of claim 1, wherein the parallel interface is an optical interface and each channel is a pixel.

6. An apparatus for data communication, comprising:

- a parallel data bus comprising a plurality of physical channels, each capable of representing a discrete state from a high-cardinality set of states;
- an encoding module configured to represent a numerical value as a single data word, wherein said data word comprises a set of parameters for generating the numerical value from a mathematical series; and

- wherein the encoding module is configured to map said data word to a single discrete state of a single physical channel.

7. The apparatus of claim 6, wherein the apparatus is configured to operate with an effective bus width of over 3.2 million times the width of a 1024-bit bus.

8. The apparatus of claim 6, wherein the parallel data bus is an 8K resolution optical display.